Binary Classification

- Model, M
- output probab. $\in [0,1]$
  - * p close to 0 – no disease
  - p close to 1 – disease
  - p close 0.5 – ambiguous
- threshold $\tau = 0.5$
  - $p \geq \tau = 0.5^+ \longrightarrow$ disease
  - $p < \tau = 0.5^- \longrightarrow$ non-disease

N = 100 Sbjs

D = 40 Sbjs

ND = 60 Sbjs

CamScanner

Confusion Matrix

Actual \ Predicting	+ve	-ve
D (40 Ref) (+ve)	35 TP	5 FN
ND (60) (-ve)	10 FP	50 TN

Sensitivity (TPR)

$$Sens = \frac{TP}{TP+FN}$$

$$= \frac{35}{35+5} = \frac{35}{40}$$

$$= 7/8 = 0.875$$

CamScanner

Specificity (TNR)

$$Spec = \frac{TN}{TN + FP}$$

$$= \frac{50}{50+10} = \frac{5}{6}$$

$$= 0.833$$

Precision

$$Prec = \frac{TP}{TP + FP} = \frac{35}{35+10} = \frac{35}{45} = \frac{7}{8}$$

$$= 0.778$$

$$Recall = \frac{TP}{TP + FN} = Sens$$

$$= 0.875$$

CamScanner

F_1 Score $= 2\,\frac{P \times R}{P + R} = \frac{2 \cdot 0.778 \cdot 0.875}{0.778 + 0.875} = 0.8235$

(Harmonic mean of Prec, Recall)

$H(a,b) = \frac{2ab}{a+b}$

$F_1 \text{ score} \in [0, 1]$

$P = \frac{TP}{\quad}$

FPR = 1 − TNR = 1 − Specificity

$= \frac{FP}{FP+TN} = \frac{10}{10+90} = \frac{1}{6} = 0.166$

For every τ

TPR = Sens = Recall

(FPR, TPR)

$$Acc = \frac{TP+TN}{TP+FP+TN+FN}$$

$$= \frac{35+8}{100}$$

$$= 0.43$$

$$Acc = \frac{TP+TN}{100}$$

$\therefore TP = 0$

$TN = 90$

$$= 0.90$$

AIP

0 TP	10 FN
5 FP	90 TN

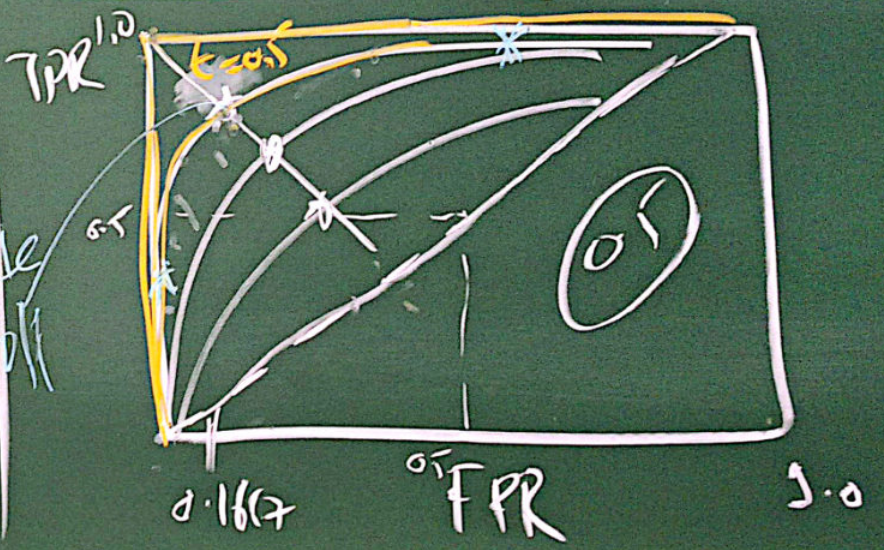
Live (+ve) -ve (97)

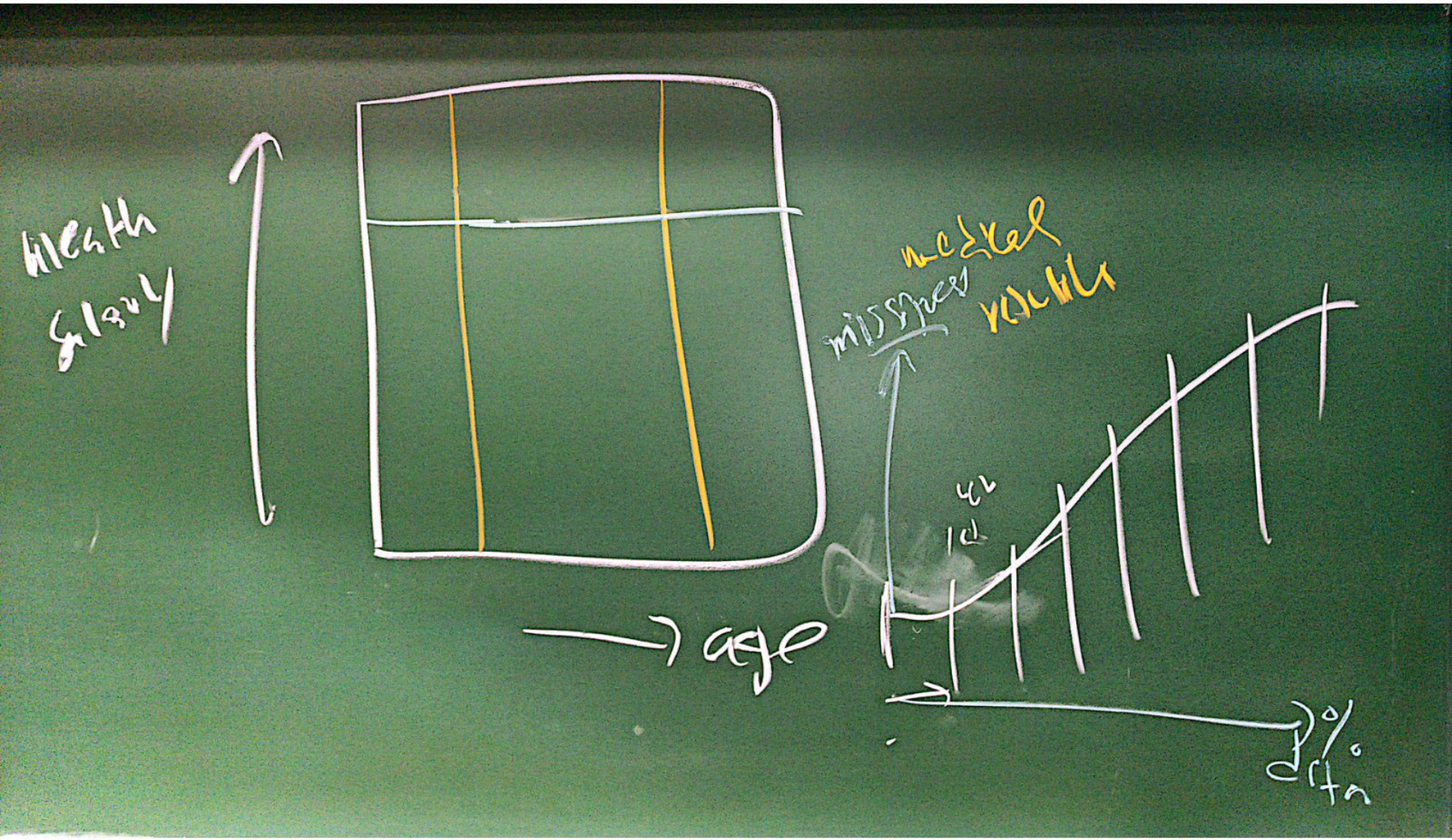
$$P = \frac{TP}{TP+FP} = 0$$

$$R = \frac{TP}{TP+FN} = 0$$

$$F_1 = \frac{2 \cdot 0 \cdot 0}{0+0} = ?0$$

AUC or AUROC max AUC = 1




Height
Stunting
age